

- Lab 3 Revis
- Virtual Mem
 - ↳ Multi-Level Page Tables
 - ↳ Addresses in x86
 - ↳ TLB
- Lab 4 discussion/question

Agenda

Li Page Table

VA

PA

36 bits	12 bits
VPN	offset

Virtual
Page Num

48-bit address
space.

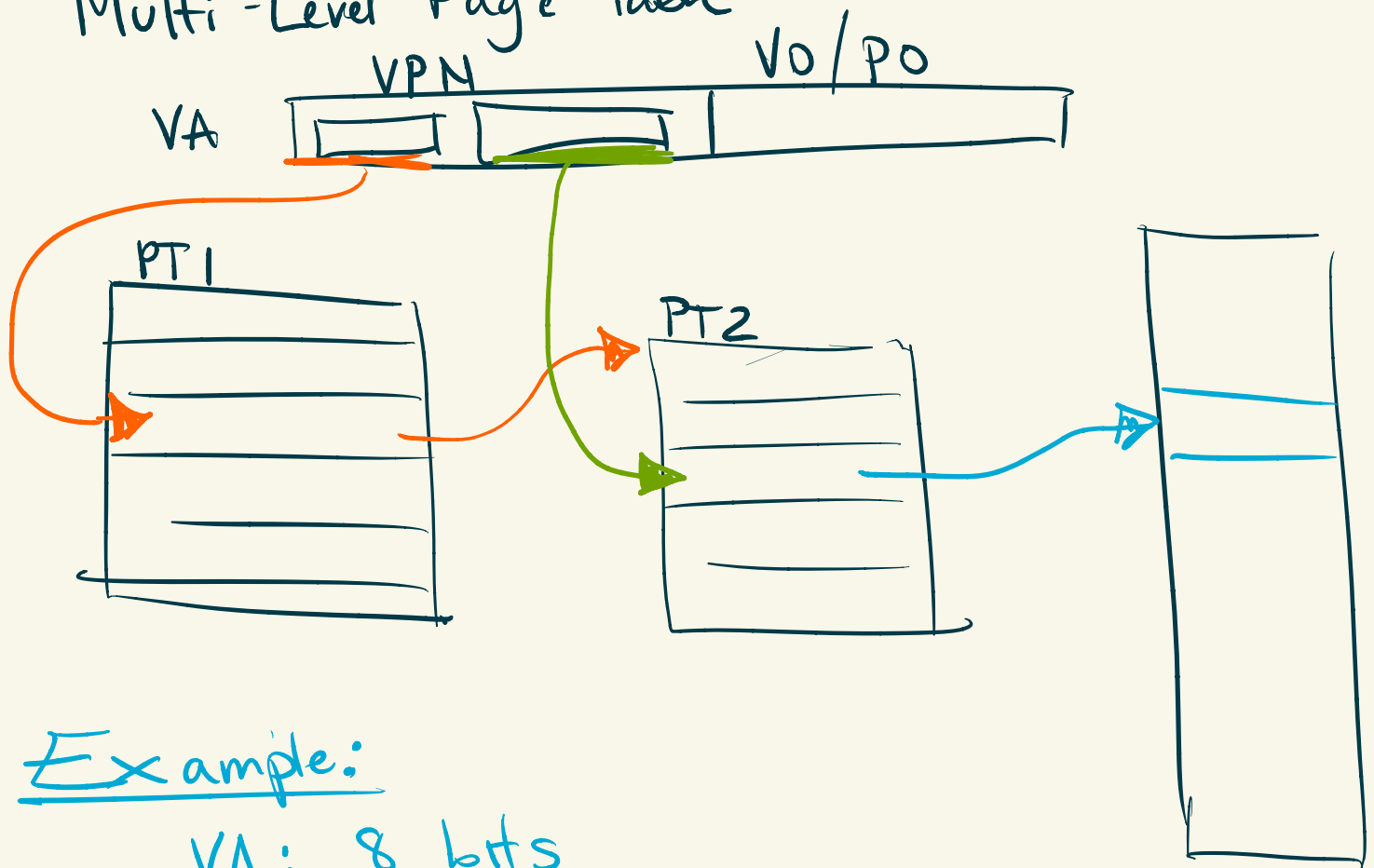
$$2^{12} = 4K \text{ page size}$$

8 byte PTE

How many pages
in the PT?

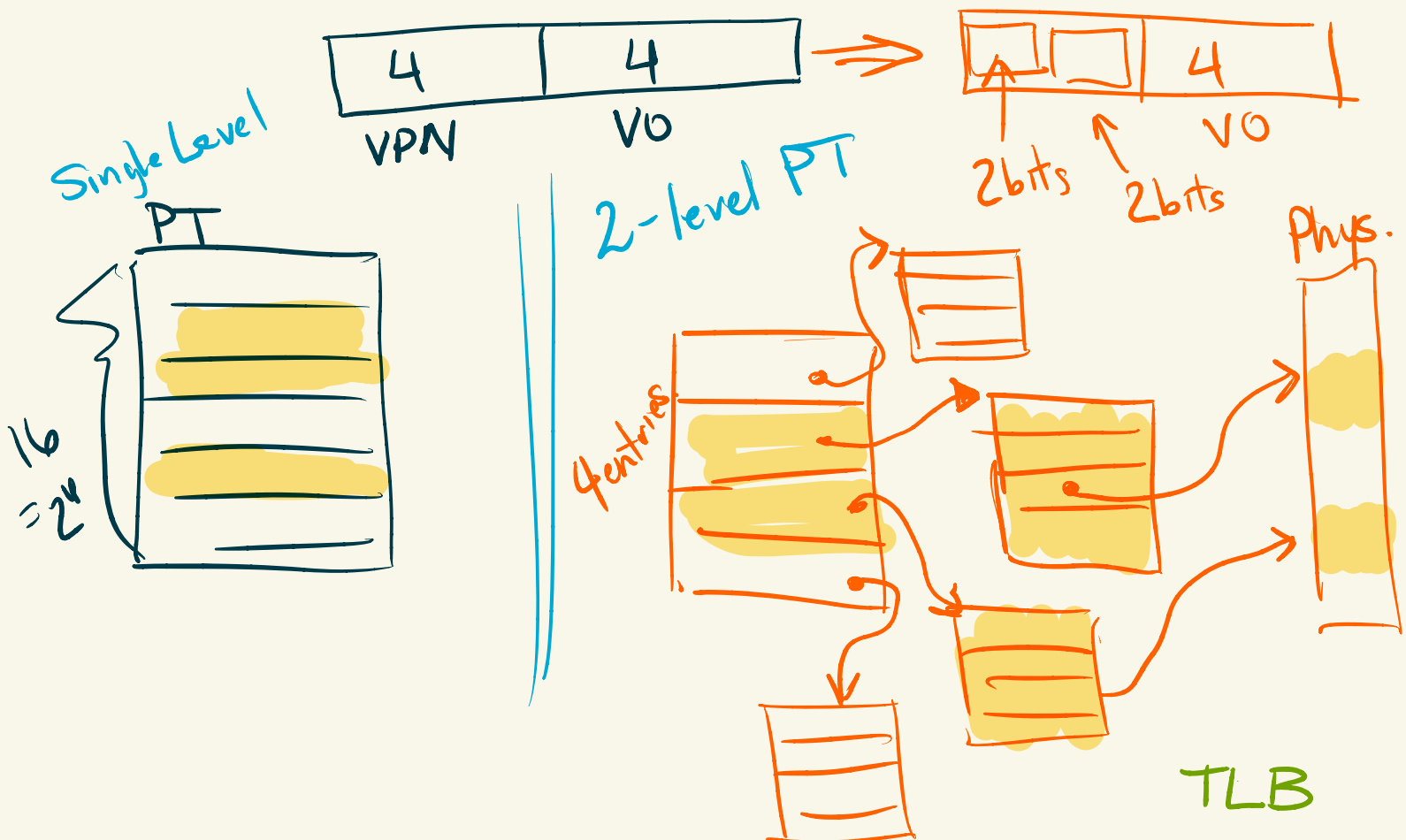
$$\begin{aligned}
 2^{36} \cdot 2^3 &= 2^{39} \\
 &= 2^{30} \cdot 2^9 \\
 &= 512 \text{ GB}
 \end{aligned}$$

Multi-Level Page Table



Example:

VA: 8 bits



Problem:

- 64 bit machine
- 16 TB RAM
- 4k page size
- 2 level PT
- Level 1 PT: 16 - 4k pages
8 byte PTEs.

Q1. How many entries can the PT contain?

$$16 \cdot 4k = 2^4 \cdot 2^{12} = 2^{16}$$

$$2^{16} / 2^3 = 2^{13} \text{ entries}$$

Q2. How many bits are needed for the L1 page table?
(in VA)

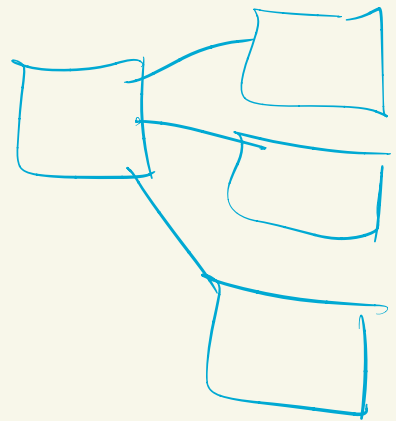
13 bits

Q3. If 34 bits are used to index into the L2 pg,
how large is the VA space?

- 13 bits for L1
- 34 bits for L2

$$2^{13} \cdot 2^{34} = 2^{47} \Leftarrow \# \text{VPNs.}$$

$$2^{47} \times 2^{12} = 2^{59} \text{ addresses}$$



x86-64 Addresses

- Virtual Addresses: 48 bits.
- Physical Address: 52 bits

